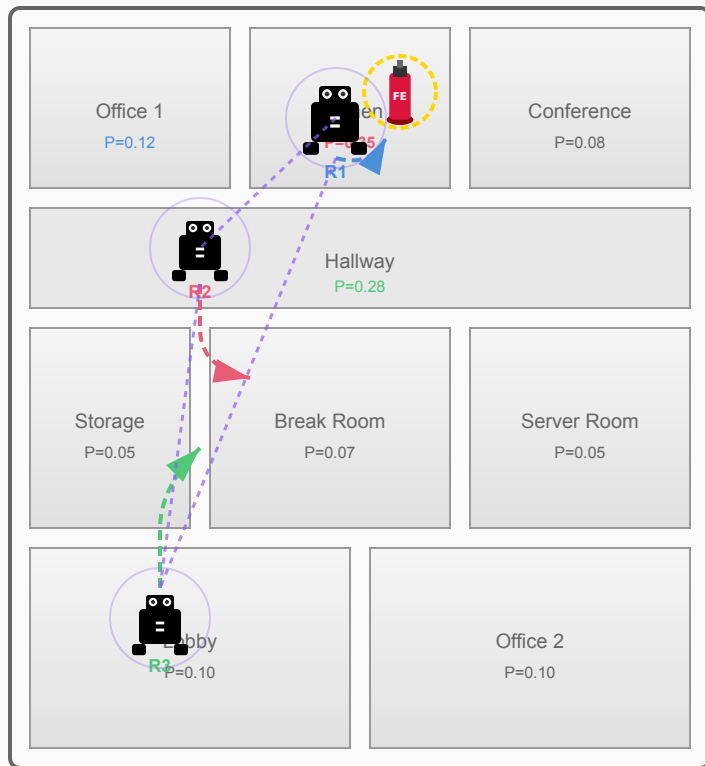
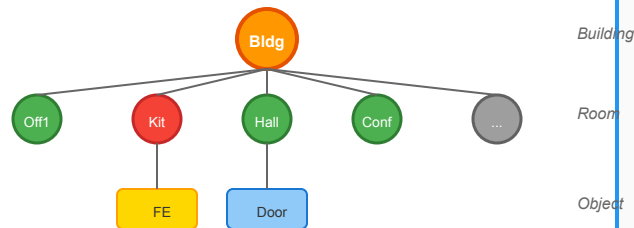


SEEK-Multi: Collaborative Multi-Agent Semantic Search

Indoor Environment

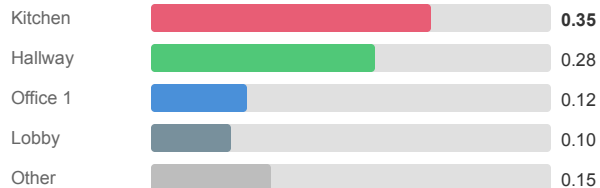


Shared Dynamic Scene Graph (DSG)



Distributed RSN Belief State

Target: Fire Extinguisher



✓ Beliefs synchronized across all agents (consensus achieved)

Legend:

Robot 1

Robot 2

Robot 3

Target Object

Communication

Planned Path

P = Probability of target in room (from RSN semantic priors)

Robots share beliefs and coordinate via distributed protocols